BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: Paper – XI (01) (23) · PR XI (02) 23

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 20 | **Paper Code:** Paper – XI (01) (23) · PR XI (02) 23

Note: Instructions (printed in Urdu on the sheet): 1. Each question has four circles; fill in only the circle of the correct answer. 2. Use a blue or black pen to fill the circles. 3. Filling more than one circle will be treated as a wrong answer.

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. In rolling a die, the probability of less than one, is.....

(A) 0 (B) 1

(C) 1/2 (D) 1/6

Q2. Amplitude of cos 2θ is equal to:

(A) 2 (B) 1/3

(C) −1/3 (D) 1

Q3. Frequency of sin 6θ is equal to:

(A) 3/π (B) π/3

(C) π/4 (D) 2π/3

Q4. tanθ is positive in.....quadrant.

(A) 1st and 2nd (B) 1st and 3rd

(C) 1st and 4th (D) 2nd and 3rd

Q5. (n−1 choose 1) + (n−1 choose 2) + (n−1 choose 3) + ... + (n−1 choose n−1) = where n > 1

(A) $2^n - 1$
(B) 2^{n-2}
(C) $2^{n-1} - 1$
(D) 2^{n-1}

Q6. The linear inequalities involved in the problem are called.....

(A) Problem constraints
(B) Feasible solution
(C) Decision variable
(D) Optimal solution

Q7. Exact value of: sin 42° cos 12° − cos 42° sin 12° =

(A) −1/2
(B) 3/2
(C) 1/2
(D) 2/3

Q8. tan(2π − θ) =

(A) tanθ
(B) cotθ
(C) −cotθ
(D) − tanθ

Q9. sin θ/2 = √((1 − cosθ)/2) where θ/2 lies in:

(A) 1st quadrant
(B) 2nd quadrant
(C) 3rd quadrant
(D) Both A and B

Q10. √(S(S − b)/(a·c)) =

(A) sin α/2
(B) cos α/2
(C) cos β/2
(D) sin β/2

Q11. √−2 × √−3 =

(A) √6
(B) −√6
(C) √−6
(D) −√−6

Q12. If Z = x + iy then which of the given is always true?

(A) $Z \cdot \bar{Z} = |Z|^2$
(B) |Z| = −|Z|
(C) $\bar{Z} = -Z$
(D) None of these

Q13. If A and B are two matrices then $(A + B)^t$ =

(A) A + B
(B) $A^t \cdot B^t$
(C) AB
(D) $A^t + B^t$

Q14. If A = [[α, 1], [1, α]] and |A| = 24 then value of α is:

(A) 5
(B) ±5
(C) 4
(D) ±4

Q15. If $\hat{u}$ is a unit vector then $|\hat{u}|$ =

(A) -1 (B) 1

(C) ± 1 (D) 2

Q16. If the nth term of an A.P is 4n + 1, then common difference is.....

(A) 3 (B) 5

(C) 4 (D) 6

Q17. Harmonic mean between two numbers a and b is.....

(A) 2ab/(a + b) (B) (a + b)/2ab

(C) (a·b)/(a + b) (D) (a + b)/(a·b)

Q18. The nth term of the series 1.2 + 2.3 + 3.4 + ...is:

(A) $n^2 + n$ (B) $n^3 - n$

(C) n^2 (D) None of these

Q19. The transpose of a row matrix is:

(A) Row matrix (B) Column matrix

(C) Square matrix (D) Unit matrix

Q20. 10th term of the sequence, 2, 5, 8, ... is:

(A) 27 (B) 31

(C) 29 (D) 33

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 50

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Answer any TEN parts. Each part carries FIVE marks.

Q1. Find the multiplicative inverse of (7, −9).

Q2. Factorize: $P(z) = z^2 + 4$

Q3. If A = [[1, 2, 3], [6, 5, 4], [4, 3, −2]] then show that $A + A^t$ is symmetric matrix.

Q4. Find the rank of the matrix if A = [[3, 1, −4], [0, 2, 1], [1, −1, −2]]

Q5. For what value of C, the following vectors are coplanar? $a = \hat{i} + \hat{j} + 2\hat{k}$, $b = 2\hat{i} + 3\hat{j} + \hat{k}$ and $c = c\hat{i} + 2\hat{j} + 6\hat{k}$

Q6. Insert two harmonic means between 12 and 48.

Q7. Show that $\hat{i} + 7\hat{j} + 3\hat{k}$ is perpendicular to both $\hat{i} - \hat{j} + 2\hat{k}$ and $2\hat{i} + \hat{j} - 3\hat{k}$

Q8. Find the sum of all the integers lying between 100 and 600 that end in 5.

Q9. Find the sum of the series: Σ (k=2 to 5) $1/(k^3 - k)$

Q10. Solve for n, if ${}^{n}P_5 = 9({}^{n-1}P_4)$

Q11. Find the inverse of the function $g(x) = 4 + \sqrt{2x}$

Q12. Graph the inequality $2x - 3y > 6$

Q13. Find the domain, range and period of $\sec(\pi/4)x$.

SECTION-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 30

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. Each question carries equal marks.

QIII.

(a) If $y = 3/4 + 3.5/(4.8) + 3.5.7/(4.8.12) + \ldots\ldots$ then show that $y^2 + 2y - 7 = 0$

(b) Prove that $\sin(\alpha + \beta)/(\cos\alpha \cos\beta) = \tan\alpha + \tan\beta$

QIV.

(a) Find the area of the inscribed circle of the triangle with measures of the sides 55m, 25m and 70m.

(b) Prove that $(\sin\alpha + \cos\alpha)/(\cos\alpha - \sin\alpha) + (\sin\alpha - \cos\alpha)/(\cos\alpha + \sin\alpha) = 2\tan 2\alpha$

QV.

(a) Find the total number of diagonals of a regular hexagon.

(b) Given that $f(x) = x^3 - ax^2 + bx + 1$ if $f(2) = -3$ and $f(-1) = 0$, then find the value of a and b.

QVI.

(a) Find maximum and minimum values of the function, $y = 1/(19 - 10\sin(3\theta - 45))$

(b) Find the area of the triangle whose vertices are A(2, 2, 0), B(−1, 0, 2) and C(0, 4, 3)